The Exact Limiting Distribution of the Dickey–Fuller t-Ratio

MATLAB programs: user manual

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Abstract

This manual documents the MATLAB programs that accompany Bao et al. (2026a) ("the paper"). The programs compute the limiting distribution function, density, quantiles and moments of the least squares t-ratio $\hat{\tau}_0 = (\hat{\phi} - \phi_0)/\operatorname{se}(\hat{\phi})$ in a Gaussian first-order autoregression under the local alternatives $\phi = 1 + c/n$, for the regression with no deterministic terms, an intercept, or an intercept and a linear trend, for the null value $\phi_0 = 1 + c_0/n$ ($c_0 = 0$: the Dickey–Fuller tests), and with a zero, a fixed or a stationary initial observation of the order of $\sqrt{n}$. Two exact representations are implemented, the single integral of Theorem 2 of the paper and the double integral of Theorem 3, and they check each other. Section 1 is a quick start, Section 2 describes the user-level routines, Section 3 the two engines, Section 4 accuracy and cost, Section 5 the validation suite and the reproduction of the paper, and Section 6 lists the files.

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1 Quick start

Add the folder `public` to the MATLAB path. The model is

$$y_t = \alpha + \delta t + \phi y_{t-1} + \sigma e_t, \qquad t = 1, \ldots, n, \qquad e_t \sim N(0,1) \text{ i.i.d.},$$

with $\phi = 1 + c/n$, no drift, and $y_0 = \sigma\sqrt{n}\,\gamma_0$. The case code names the start (Deterministic, that is fixed, or Stationary) and the regressors (intercept Known to be zero; Unknown intercept; intercept and Trend):

Code	Regression	Start
`'DK'`	y_t on y_{t-1} ($\hat{\tau}$)	γ_0 fixed (`opts.g0`, default 0)
`'DU'`	y_t on 1, y_{t-1} ($\hat{\tau}_\mu$)	γ_0 fixed
`'DT'`	y_t on 1, t, y_{t-1} ($\hat{\tau}_\tau$)	γ_0 fixed
`'SK'`, `'SU'`, `'ST'`	the same	stationary, $J(0) \sim N(0, -1/(2c))$, $c < 0$

All routines take `(...,cas,c,opts)`: the case, the local-to-unity parameter c (default 0, the unit root) and an optional structure of options, of which `c0` (the local null value, default 0), `g0` (the fixed start, default 0) and `method` (`'auto'`, the default, `'single'` or `'double'`) are the ones a user needs.

```
ltratio_icdf([0.01 0.05 0.10], 'DU')        % asymptotic critical values of tau_mu
ltratio_cdf(-2.86157, 'DU', -5)              % limiting power of the 5% test at c =
    -5
ltratio_cdf(-2.86157, 'SU', -5)              % the same with a stationary start
ltratio_cdf(-2.86157, 'DU', -5, struct('g0', 1.5))   % fixed start y_0 = 1.5 sigma
    sqrt(n)
ltratio_cdf(0, 'DK', 2, struct('c0', 2))     % t-ratio centered at the true value, c
    = 2
ltratio_pdf(-1:0.5:1, 'DK')                  % the density of tau at five points
ltratio_mom(4, 'DT')                         % E tau_tau^j, j = 1..4
```

The first call returns -3.43035, -2.86157 and -2.56679; the second 0.1174. A value of the distribution function or the density takes a fraction of a second, a quantile a few values, and a set of moments about a second.

2 The user-level routines

Common arguments. `cas` is the case code (in either letter case), `c` a finite real scalar (negative for a stationary start), and `opts` a structure. The arguments are checked by `ltratio_args`, which rejects unknown codes, a stationary start with $c \geq 0$, a fixed start `g0` in a stationary case and an unknown `method`. Fields of `opts` other than `c0`, `g0` and `method` are passed to the engine in use (Section 3). Vector arguments are evaluated elementwise.

`[F,info] = ltratio_cdf(x,cas,c,opts)`
The limit of $\mathbb{P}[\hat{\tau}_0 \leq x]$. At `x = -Inf`, `Inf` and `NaN` the result is 0, 1 and `NaN`. `info.method` is the representation used and `info.time` the elapsed time; with `'double'` the diagnostics

of `tr_limdist` are added (`info.status`, `info.err`, ...; a nonzero status raises the warning `tr_limdist:status`).

`[f,info] = ltratio_pdf(x,cas,c,opts)`
The limiting density; arguments and options as for `ltratio_cdf`. In case DK with $c_0 > 0$ the density is infinite at $x = -\sqrt{2c_0}$ (a logarithmic singularity, where the two zeros of the kernel's edge merge) and `Inf` is returned there.

`[q,info] = ltratio_icdf(p,cas,c,opts)`
The limiting p-quantiles, $p \in (0,1)$, by Newton's method on `ltratio_cdf` with `ltratio_pdf` as derivative, safeguarded by bisection, started at the normal approximation from the first two moments. `opts.tol` (10^{-10}) is the tolerance on q; `info.F` is the distribution function at q, `info.iter` the iteration count and `info.conv` the convergence flag (a failure raises `ltratio:icdf`).

`[m,info] = ltratio_mom(p,cas,c,opts)`
The moments $\mathbb{E}(\tau^j)$, $j = 1,\ldots,p$, of the limit $\tau = (U - c_0V)/\sqrt{V}$, as one-dimensional integrals of the joint moment generating function at real arguments (Proposition 3 of the paper), through `tr_limmom`; the options `c0` and `g0` apply, `method` does not. `info` holds the error estimate of the integral per moment and the time.

The two representations. With $U = \int_0^1 \tilde{J}\mathrm{d}J$ and $V = \int_0^1 \tilde{J}^2\mathrm{d}r$, J the Ornstein–Uhlenbeck process $\mathrm{d}J = cJ\mathrm{d}r + \mathrm{d}W$ started at γ_0 and $\tilde{J}$ its residual on the deterministic terms, the limit is $\tau = (U - c_0V)/\sqrt{V}$, and the joint moment generating function of (U,V) is $e^{-s/2}D(s,t)^{-1/2}$ with D a polynomial of degree one (DK) or two in s (Proposition 1 of the paper), to whose logarithm a fixed start adds $\gamma_0^2 Q(s,t)$ in closed form (Proposition 2). The *single integral* (`method = 'single'`, Theorem 2) inverts in s explicitly, which leaves a scaled χ^2_1 density (DK) or a Bessel K_0 density of $U + \frac{1}{2}$ under the tilt e^{tV}, and integrates its Gaussian transforms along the line $\mathrm{Re}\, t = \tau$; its engine is `dfx_core`. The *double integral* (`method = 'double'`, Theorem 3) keeps the outer integral over the value v of V and inverts in t, for each v, along a path through a saddle point; its engine is `tr_limdist`. A fixed start $\gamma_0 \neq 0$ in cases DU and DT makes the tilted density noncentral and without a closed form, and only the double integral is available there; `ltratio_cdf` and `ltratio_pdf` then use it whatever `method` says. Everywhere else both are available and agree to about 10^{-12} in the distribution function (`ltratio_test`, Section 5). With `method = 'auto'`, the default, `ltratio_method` chooses point by point the representation that is the faster one in these programs, by their own timings (`validation/method_sweep.m`): the single integral at $c_0 = 0$, where it is 1.2 to 60 times faster; with $c_0 \neq 0$ the double integral in cases DK, DU, DT and SK, except for $c_0 > 0$ at $x \geq 0$, where the single integral has no breakpoint and is 3 to 30 times faster (with $c_0 < 0$ its saddle factor costs up to 11 seconds at x of 4 to 8, while the double integral stays below half a second); and the single integral in cases SU and ST, whose double integral costs half a second a value, except for $c_0 < 0$ at $x \geq 0$. The C and Julia implementations apply the same rule for every start; the exception for cases SU and ST at $x < 0$ is the one place where the MATLAB timings differ.

3 The two engines

The single integral: `dfx_core`. Called through `dfx_cdf(q,model,c,opts)` and the front ends `dfx_pdf`, `dfx_quantile`, `dfx_moments`, `dfx_tstar`, with the model number 1, 2, 3 for K, U, T and `opts.start` equal to `'zero'` (default) or `'stationary'`; `ltratio_dist` does the mapping. The port follows the Julia implementation `DFExact.jl` of the single integral function by function. The line is $\operatorname{Re} t = \tau$ with $\tau = 1$ for $c \le 0$ and a zero start and $\min(1, t_*/2)$ otherwise, t_* the convergence abscissa of the moment generating function of V (`dfx_tstar`; located from the cancellation-free Girsanov forms for $c > 2$, so that c up to 20 and beyond is reached). The integrand is split, by an entire partition of unity inside the inner transforms, into a smooth part, integrated by the trapezoidal rule in $w = \tau \sinh u$ with the step 0.075 up to $w = 4900$, and oscillating parts $G_j(w)e^{-\mathrm{i}wv_j^2}$, one for each positive root v_j of $X(v) = c_0v^2 + xv + \frac{1}{2}$ (one root, $1/(2|x|)$, at $c_0 = 0$ and $x < 0$), integrated by Filon's method on geometric panels with 20 Gauss–Legendre nodes; for $c > 0$, and in the upper tail with $c_0 < 0$, the smooth part carries a saddle-point factor whose phase grows like x^2, and it is integrated on Gauss–Legendre panels whose node counts follow the phase. The inner transforms are Gauss–Legendre integrals on complex paths on which the Gaussian kernel decays; with $c_0 \neq 0$ the paths between the breakpoints (the roots of X, and the vertex of the parabola for $c_0 > 0$) are straight pieces along the directions of steepest descent, joined through the real axis where a direct connection would cross a region in which the Gaussian grows (Appendix B.1 of the paper). Options (for checking): `tau` (the line), `hu` (0.075), `N` (20), `pfac` (subdivides the Filon panels), `glfac` (multiplies the node counts of the Gauss–Legendre route), `wmax` (4900), `gen` (`true`: the general kernel also at $c_0 = 0$), `uniform` (the trapezoidal rule for the whole integrand); `dfx_core('nodefac',2)` doubles every node count until reset with `1`. The moments of `dfx_moments` use the form of Proposition 1 and lose accuracy for $c > 2$; `ltratio_mom` uses `tr_limmom`.

The double integral: `tr_limdist`. `[F,info] = tr_limdist(x,cas,c,opts,mode)`, `mode` `'cdf'` or `'pdf'`, with `opts.c0` (default c there, not 0: `ltratio_dist` passes `c0` explicitly) and `opts.g0`. In case DK it uses part (i) of Theorem 3, whose inner probability is an error function (with a fixed start, the probability $\mathbb{P}[X^2 \le 2a + \gamma_0^2]$ of a noncentral normal); with a zero start in cases DU, DT and with a stationary start in all three cases part (ii), whose integrands are in closed form; and in cases DU, DT with a fixed start part (i) with the noncentral probability (`Pnc`), a one-dimensional integral with `N1` (96) Gauss–Legendre nodes, so a triple integral in all. With a zero start the integrands are the Girsanov forms (`iipar`), analytic beyond t_*, and the inner inversions run through the saddle point of the whole integrand on the better of two paths (a parabola turning into a wedge; a vertical line in the ϑ-plane), with 96 trapezoidal nodes after a sinh substitution, doubled up to three times until the nested-rule estimate is below 10^{-13}. With an initial observation the path is the Talbot parabola through the saddle point of $\log M_V(t) - tv$, with the sign conditions of Theorem 3 checked at every node (`info.nbad` counts violations) and the branch of $\log D$ continued along it; `Nv` (96) nodes, doubled up to twice where the nested estimate exceeds 10^{-9} of the largest column of the piece and 10^{-11} divided by the range of v (`info.nunres` counts the columns still unresolved). The initial-value terms come from `tr_nsstart` (Proposition 2: $Q(s,t)$ in closed form, from Nabeya and Sørensen's determinants in cases DU, DT and the Riccati solution `tr_jbstart` in case DK); the squared means of the noncentral form are read off Q at three values of s. The outer integral over v is a cosine-mapped Gauss–Legendre rule in $\log v$ over $[e^{-8}, e^{6}]\,\mathbb{E}V$ with `M` (64; 32 in case DK with a zero start) nodes

per part of length at most `plen` (2), breakpoints at the zeros of $a(v) = \frac{1}{2} + x\sqrt{v} + c_0 v$, grading toward the step of the integrand in the far upper tail for $c > c_0$ (width below `stepw`, 0.1) and bisection, up to `maxlev` (12) times, where the Legendre-coefficient estimate exceeds `otol` (10^{-12}). `info.err` is the error estimate per point and `info.status` is 0 when clean, otherwise 1 (sign violations) plus 2 (unresolved inner integrals, or `info.err` above `errtol`, 10^{-10}); a nonzero status raises the warning `tr_limdist:status`. Further options: `blo`, `bhi` (the range in $\log v$ relative to $\log \mathbb{E}V$), `region` (checks the sign conditions on that many intermediate paths), `oldDK`, `oldII` (the earlier forms through the saddle point of M_V alone, which fail for c above about 2 and 5).

The moment generating function. `tr_cflim(t1,t2,cas,c,form)` evaluates the Nabeya–Tanaka determinants of Proposition 1 at complex arguments (31-term Taylor series in z for $|z| < 4$ from `nt_series`, closed forms in $\vartheta = (-z)^{1/2}$ and $e^{-2\vartheta}$ beyond), with the branch of $\log D$ followed along two legs and the roots of D in s (the weights κ_k of $U + \frac{1}{2}$ under the tilt); the forms `'F'`, `'B'`, `'BC'` and `'ApB'` return the coefficients, the weights, the branch-continued logarithm along a path, and the moment generating function of U and V together. `tr_limmom(p,cas,c,opts)` computes the moments by Cauchy integrals on a circle in the argument of U and the trapezoidal rule in $\log s$, from the Girsanov forms with a zero start and with the initial-value terms of `tr_nsstart` otherwise.

4 Accuracy and cost

Both representations are accurate to about 10^{-11} in the distribution function and 10^{-10} relative in the density over $c \in [-30, 20]$, $c_0 \in [-40, 10]$ and the six cases, and to 10^{-11} with a fixed start in cases DU, DT for $c \le 2$ (Section 7 and Appendix B of the paper). The error of the single integral is absolute rather than relative: its terms are of order one, and it carries an error of 10^{-14} to 10^{-21} whatever the size of the value, which finer rules do not reduce (2×10^{-14} in the lower tails of cases DT and ST with $c_0 \neq 0$). When `opts.method` is not given, `ltratio_pdf` and `ltratio_cdf` therefore recompute by the double integral every value of the single integral below 10^{-3} (a density, or a distribution function in the lower tail), with its error estimate in `info.err`. The relative accuracy then holds in the tails as well, down to that estimate; far out, below the error of either method, a computed density can be a tiny negative number, and it is returned as zero. Doubling every node count of the single integral changes the distribution function by at most 4×10^{-13}; the two representations agree to 4×10^{-12} at worst (Table 1 of the paper, reproduced by `validation/acc_table.m`). With a fixed start in cases DU, DT at c above about 2.5 the Talbot path of the double integral no longer satisfies the sign conditions everywhere, and `tr_limdist` warns; the single integral does not cover that configuration. A value of the distribution function or the density takes, in MATLAB on one thread of a laptop, 0.1 to 0.3 seconds by the single integral for $c_0 = 0$ and 0.3 to 5 seconds with $c_0 \neq 0$ (the upper end in the upper tail at $c = 5$, $c_0 = -5$), and 0.05 to 0.4 seconds by the double integral with a zero start and 0.3 to 1.3 seconds with an initial observation; a quantile takes a few values and a set of four moments about a second. The Julia implementation of the single integral, `DFExact.jl`, takes 7 to 17 milliseconds per value on six threads.

5 Validation and reproduction of the paper

`ltratio_test` runs a hundred and fourteen checks in about two minutes and reports a failure count: the two representations against each other at the mean plus -2.5 to 2.5 standard deviations in the six cases, at $c \in \{0, -5, -10\}$, $c = 2$ and 5 and $c_0 \neq 0$; the exact values $\mathbb{P}[\hat{\tau} \leq 0] = \mathrm{erf}\{(2w)^{-1/2}\}$, $w = (e^{2c} - 1)/(2c)$, for $c \in \{-10, 0, 2, 5, 20\}$, $\mathbb{P}[\hat{\tau}_\mu \leq 0] = 0.95756559$ from the coefficient statistic, and $\mathbb{P}[U - c_0 V \leq 0]$ at $c = 10$, $c_0 = 5$ from an independent one-dimensional inversion; the distribution function at the asymptotic critical values of MacKinnon (2010) and the quantile function; the initial observation (continuity as $\gamma_0 \to 0$; the stationary start as the 24-point Gauss–Hermite average of the fixed start; the noncentral probability $\mathbb{P}[J(1)^2 \leq 1 + \gamma_0^2]$ in case DK; the rank-two structure of the initial-value term); the moments against Nabeya (1999), against `dfx_moments` and against the integral of the density; the density against fifth-order differences of the distribution function; and the argument checks. `ltratio_paper` recomputes the tables of the paper and prints each value next to the published one (about ten minutes); `validation/acc_table.m` produces the accuracy table and its LaTeX rows.

6 Files

`ltratio_cdf.m, ltratio_pdf.m, ltratio_icdf.m, ltratio_mom.m`
: the user-level routines; `ltratio_dist.m` (dispatch to the engines), `ltratio_method.m` (the automatic choice of the representation) and `ltratio_args.m` (argument checks).

`dfx_core.m, dfx_cdf.m, dfx_pdf.m, dfx_quantile.m`
: the single integral: the engine and its front ends;

`dfx_moments.m, dfx_tstar.m, dfx_series.m, dfx_test.m`
: its moments, convergence abscissa, series coefficients and test.

`tr_limdist.m, tr_limmom.m`
: the double integral and the moments;

`tr_cflim.m, tr_nsstart.m, tr_jbstart.m, nt_series.m, gen_series.py, faddeeva.m`
: the moment generating function, the initial-value terms, the series coefficients and the Faddeeva function.

`ltratio_test.m, ltratio_paper.m, validation/`
: validation and reproduction.

`Contents.m, README.md, LICENSE, doc/`
: this manual.

References

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Nabeya, S. (1999). Asymptotic moments of some unit root test statistics in the null case. *Econometric Theory* 15, 139–149.